

Capture the flag 2: Widgets

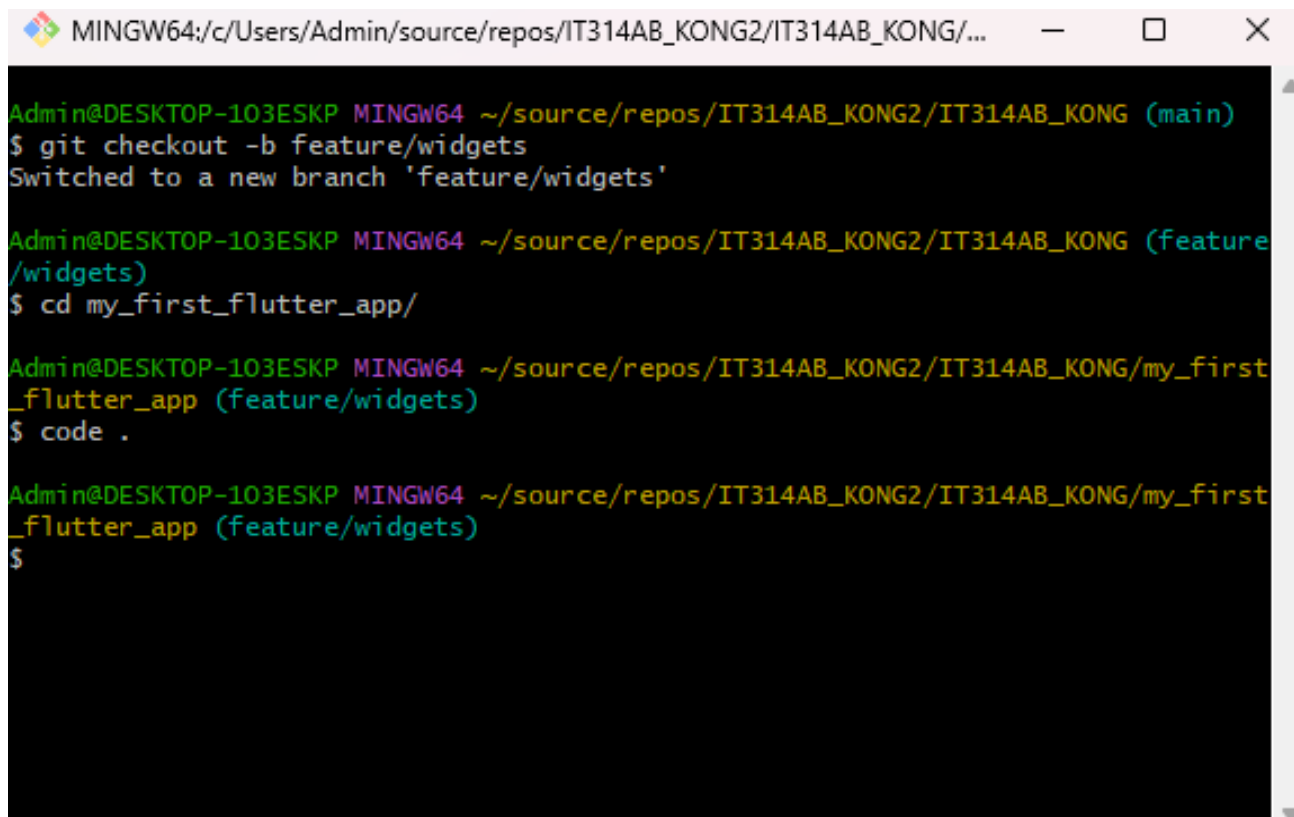
Goal: Turn the single-card application into a simple multi-section interface.

🚩 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`

A screenshot of a Windows terminal window with a black background and green and white text. The window title bar shows the path 'MINGW64:/c:/Users/Admin/source/repos/IT314AB_KONG2/IT314AB_KONG/...' and standard window controls. The terminal output shows the user 'Admin@DESKTOP-103ESKP' in the 'main' branch of a repository. They run 'git checkout -b feature/widgets', which switches to a new branch. Then they run 'cd my_first_flutter_app/' to change to the application directory. Finally, they run 'code .' to open the application in a code editor. The prompt '\$' is visible at the end of the last line.

```
Admin@DESKTOP-103ESKP MINGW64 ~/source/repos/IT314AB_KONG2/IT314AB_KONG (main)
$ git checkout -b feature/widgets
Switched to a new branch 'feature/widgets'

Admin@DESKTOP-103ESKP MINGW64 ~/source/repos/IT314AB_KONG2/IT314AB_KONG (feature
/widgets)
$ cd my_first_flutter_app/

Admin@DESKTOP-103ESKP MINGW64 ~/source/repos/IT314AB_KONG2/IT314AB_KONG/my_first
_flutter_app (feature/widgets)
$ code .

Admin@DESKTOP-103ESKP MINGW64 ~/source/repos/IT314AB_KONG2/IT314AB_KONG/my_first
_flutter_app (feature/widgets)
$
```

🚩 Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

```
Text(  
  'Chiara Marie C. Canque\n'  
  'BSIT - 3\n'  
  'My First Flutter Application\n'  
  'August 11, 2026',  
)
```

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

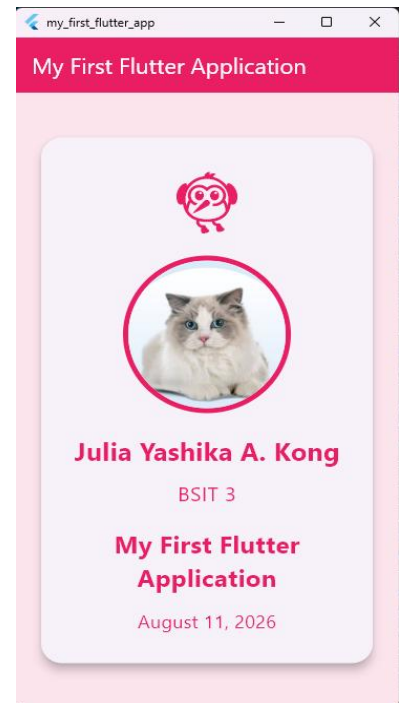
Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application

```
Text("Julia Yashika A. Kong"),  
Text("BSIT 3"),  
Text("My First Flutter Application"),  
Text("August 11, 2026"),|
```



🚩 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

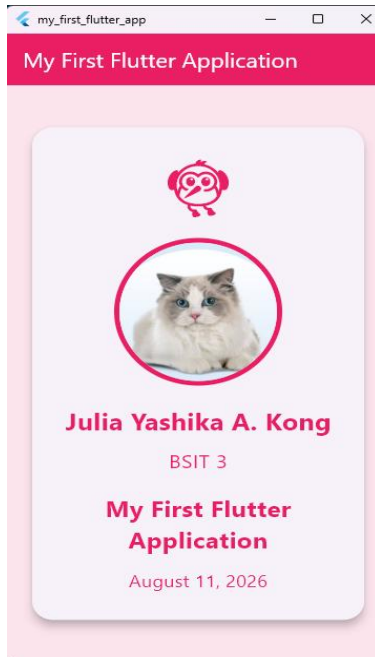
Course & Section

Application Title

Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically



```
child: Column(  
  mainAxisAlignment: MainAxisAlignment.center,  
  children: [  
    Image.asset(  
      'assets/meow.png',  
      width: 150,  
    ),  
    SizedBox(height: 20),  
    Text("Julia Yashika A. Kong"),  
    Text("BSIT 3"),  
    Text("My First Flutter Application"),  
    Text("August 11, 2026"),  
  ],  
)
```

🚩 Flag 3 – Add Another Element

Objective: Add another individual widget to your `Column`.

Add something that belongs to your profile.

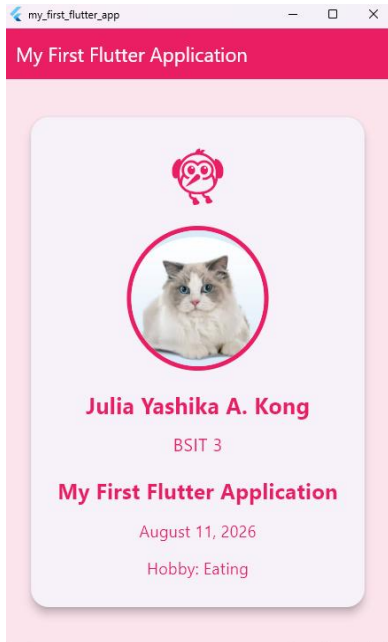
Examples:

Hobby

It must be its **own widget**, not additional text inside another `Text`.

Screenshot:

- New widget in your `Column`
- Updated application



```
child: Column(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [
    Image.asset(
      'assets/meow.png',
      width: 150,
    ),
    SizedBox(height: 20),
    Text("Julia Yashika A. Kong"),
    Text("BSIT 3"),
    Text("My First Flutter Application"),
    Text("August 11, 2026"),
    Text("Hobby: Eating"),
  ],
),
```

🚩 Flag 4 – Put Two Things Side-by-Side

Objective: Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

`Column`

`├── Text`

`├── Text`

`└── Text`

Text

Text

Row

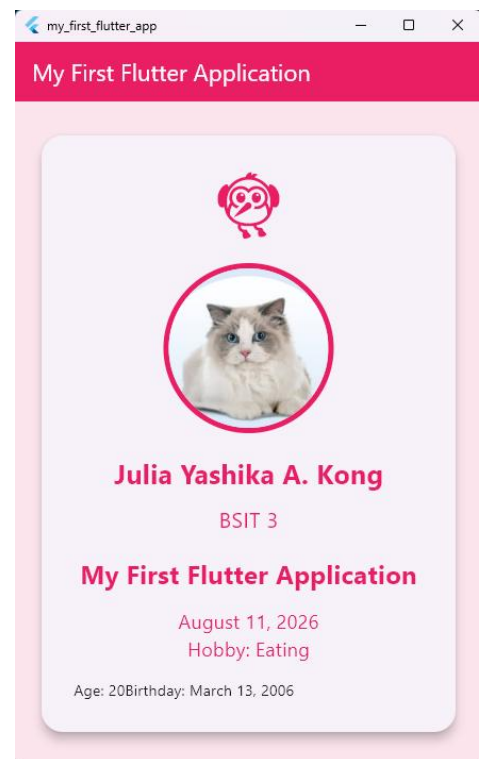
Text

Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side

```
Row(  
  children: [  
    Text('Age: 20'),  
    Text('Birthday: March 13, 2006'),  
  ],  
)
```



Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizedBox()`

Add appropriate spacing between your:

- Image
- Name
- Course

- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet.**

Screenshot:

- **SizeBox** usage
- Updated application

```

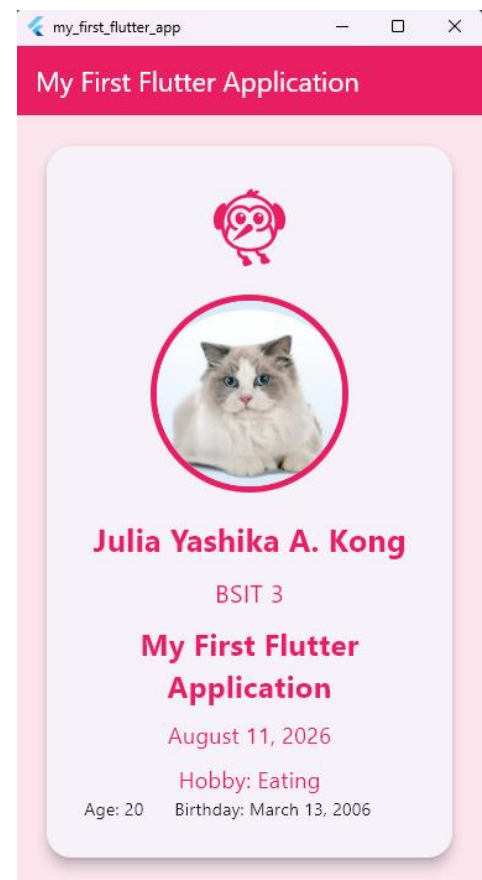
93         style: TextStyle(
94           fontSize: 24,
95           fontWeight: FontWeight.bold,
96           color: Colors.pink,
97         ),
98       ),
99     ),
100     const SizedBox(height: 10),
101   Text(
102     'August 11, 2026',
103     style: TextStyle(
104       fontSize: 18,
105       color: Colors.pink,
106     ),
107   ),
108 ),
109 const SizedBox(height: 10),
110 Text(
111   'Hobby: Eating',
112   style: TextStyle(
113     fontSize: 18,
114     color: Colors.pink,
115   ),
116 ),
117 ),
118 ),
119 Row(
120   children: [
121     Text('Age: 20'),
122     SizedBox(width: 25),
123     Text('Birthday: March 13, 2006'),
124   ],
125 ),
126 ),
127 ),
128 ),
129 ),
130 ),

```

```

67 const SizedBox(height: 20),
68 Text(
69   'Julia Yashika A. Kong',
70   style: TextStyle(
71     fontSize: 25,
72     fontWeight: FontWeight.bold,
73     color: Colors.pink,
74   ),
75 ),
76 ),
77 const SizedBox(height: 10),
78 Text(
79   'BSIT 3',
80   style: TextStyle(
81     fontSize: 20,
82     color: Colors.pink,
83   ),
84 ),
85 ),
86 ),
87 const SizedBox(height: 10),
88 Text(
89   'My First Flutter Application',
90   textAlign: TextAlign.center,
91   style: TextStyle(
92     fontSize: 24,
93     fontWeight: FontWeight.bold,
94     color: Colors.pink,
95   ),
96 ),
97 ),
98 ),
99 const SizedBox(height: 10),
100 Text(
101   'August 11, 2026',
102   style: TextStyle(

```



🚩 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets. Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

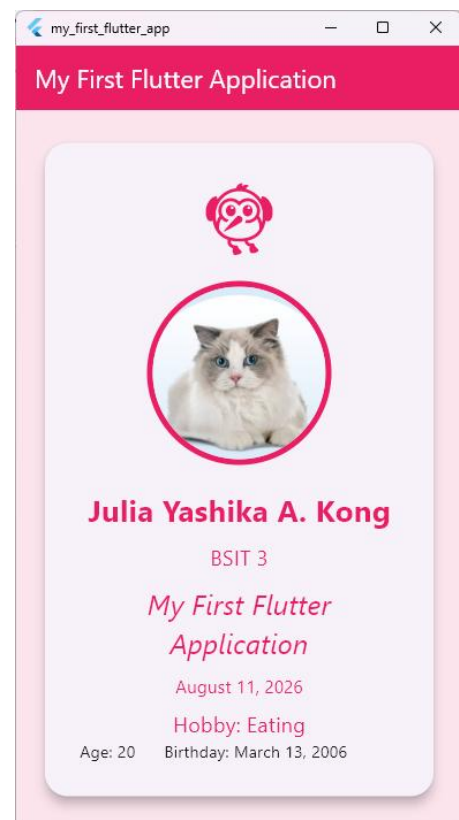
For example:

- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes
- Updated application

```
69 Text(  
70   'Julia Yashika A. Kong',  
71   style: TextStyle(  
72     fontSize: 26,  
73     fontWeight: FontWeight.bold,  
74     color: Colors.pink,  
75   ),  
76 ),  
77 ),  
78 const SizedBox(height: 10),  
79   
80 Text(  
81   'BSIT 3',  
82   style: TextStyle(  
83     fontSize: 18,  
84     color: Colors.pink,  
85   ),  
86 ),  
87 ),  
88 const SizedBox(height: 10),  
89   
90 Text(  
91   'My First Flutter Application',  
92   textAlign: TextAlign.center,  
93   style: TextStyle(  
94     fontSize: 24,  
95     fontStyle: FontStyle.italic,  
96     color: Colors.pink,  
97   ),  
98 ),  
99 ),  
100 const SizedBox(height: 10),  
101   
102 Text(  
103   'August 11, 2026',  
104   style: TextStyle(  
105     fontSize: 15,  
106     color: Colors.pink,  
107   ),  
108 ),  
109 ),  
110 const SizedBox(height: 10),  
111   
112 Text(  
113   'Hobby: Eating',  
114   style: TextStyle(  
115     fontSize: 18,  
116     color: Colors.pink,  
117   ),  
118 ),  
119 ),
```



🚩 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

`mainAxisAlignment`

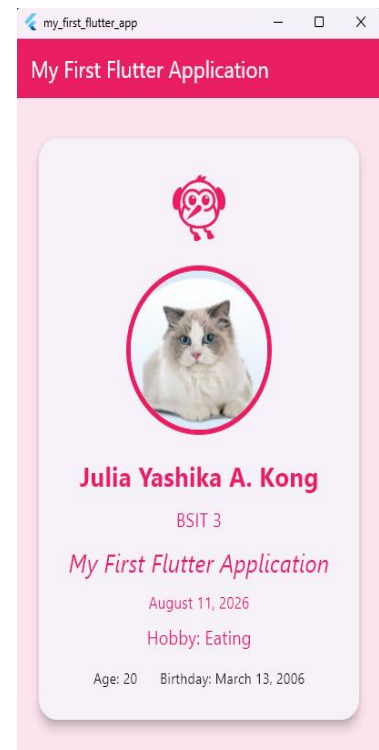
`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed
- Updated application

```
const SizedBox(height: 15),  
  
Row(  
  mainAxisAlignment: MainAxisAlignment.center,  
  crossAxisAlignment: CrossAxisAlignment.center,  
  children: [  
    Text('Age: 20'),  
    SizedBox(width: 25),  
    Text('Birthday: March 13, 2006'),  
  ],  
)
```



🚩 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second **Card** containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

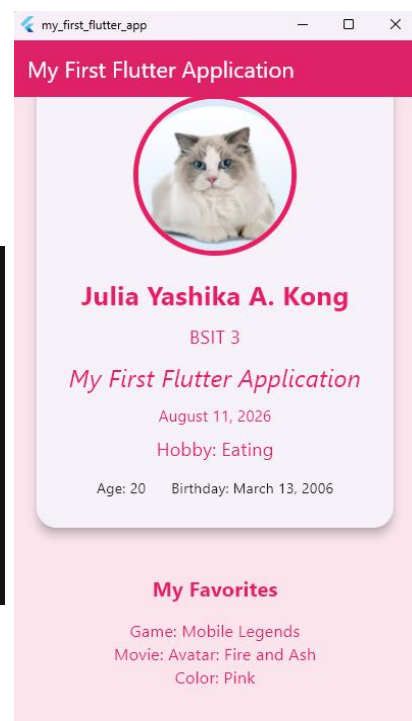
- Individual **Text** widgets
- A **Column**
- At least one **Row**
- Spacing

Screenshot:

- Code for the second card
- Both cards displayed

```
145 // FAVORITES CARD
146 Card(
147   color: Colors.pink.shade50,
148   elevation: 0,
149   child: Padding(
150     padding: const EdgeInsets.all(16.0),
151     child: Column(
152       mainAxisAlignment: MainAxisAlignment.center,
153       children: [
154         Text(
155           'My Favorites',
156           style: TextStyle(
157             color: Colors.pink,
158             fontSize: 20,
159             fontWeight: FontWeight.bold,
160           ),
161         ),
162         const SizedBox(height: 16),
163         Column(
164           mainAxisAlignment: MainAxisAlignment.center,
165           crossAxisAlignment: CrossAxisAlignment.center,
166           children: [
167             Text(
168               'Game: Mobile Legends',
169               style: TextStyle(
170                 color: Colors.pink,
171                 fontSize: 16,
172               ),
173             ),
174             const SizedBox(height: 15),
175             Text(
176               'Movie: Avatar: Fire and Ash',
177               style: TextStyle(
178                 color: Colors.pink,
179                 fontSize: 16,
180               ),
181             ),
182             const SizedBox(height: 15),
183             Text(
184               'Color: Pink',
185               style: TextStyle(
186                 color: Colors.pink,
187                 fontSize: 16,
188               ),
189             ),
190           ],
191         ),
192       ],
193     ),
194   ),
195 );
```

```
187 const SizedBox(height: 15),
188
189 Text(
190   'Color: Pink',
191   style: TextStyle(
192     color: Colors.pink,
193     fontSize: 16,
194   ),
195 ),
196 ],
197 ),
198 ],
199 ),
200 ],
201 ),
202 ),
203 ),
```



🚩 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card



Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application



🚩 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

- Icon
- Divider
- CircleAvatar
- Chip
- ElevatedButton

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application



```

184
185
186
187
188
189
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200
201
202
203
204
205

```

```

Icon(
  Icons.favorite,
  color: Colors.pink,
  size: 20,
), // Icon

Text(
  'My Favorites',
  style: TextStyle(
    color: Colors.pink,
    fontSize: 18,
    fontWeight: FontWeight.bold,
  ), // TextStyle
), // Text

 SizedBox(width: 5),

 Icon(
  Icons.favorite,
  color: Colors.pink,
  size: 20,
), // Icon

```

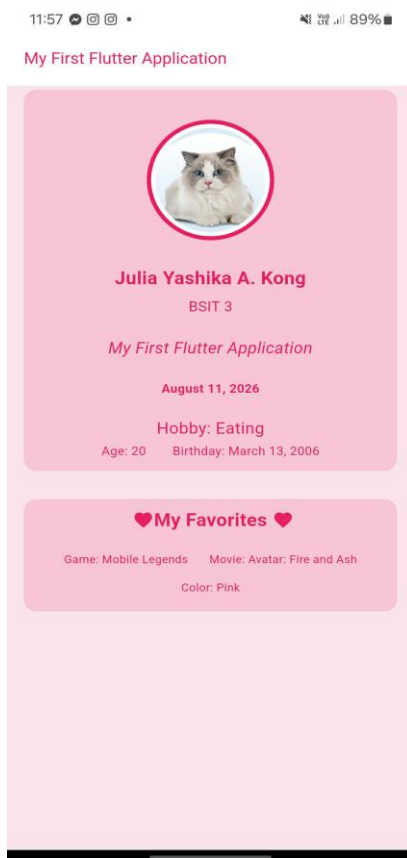
🚩 Flag 11 – Real Phone Test ☐

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)



🚩 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- `feature/widgets` branch

```
student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG2
$ ls
IT314AB_KONG/

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG2
$ cd IT314AB_KONG/

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG2/IT314AB_KONG (feature/widgets)
$ git add .
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG2/IT314AB_KONG (feature/widgets)
$ git commit -m "initial commit"
[feature/widgets 05933d7] initial commit
3 files changed, 231 insertions(+), 100 deletions(-)

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG2/IT314AB_KONG (feature/widgets)
$ git push origin feature/widgets
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 1.91 KiB | 1.91 MiB/s, done.
Total 7 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
remote:
remote: Create a pull request for 'feature/widgets' on GitHub by visiting:
remote:   https://github.com/yashipikapa/IT314AB_KONG/pull/new/feature/widgets
remote:
To https://github.com/yashipikapa/IT314AB_KONG.git
 * [new branch]   feature/widgets -> feature/widgets

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG2/IT314AB_KONG (feature/widgets)
$ |
```